

e.g. heating of bulb would affect only rate of boiling

A1 [2]

- 3 (a) use of $a = -\omega^2 x$ clear
 either $\omega = \sqrt{(2k/m)}$ or $\omega^2 = (2k/m)$
 $\omega = 2\pi f$
 $f = (1/2\pi)\sqrt{(2 \times 300)/0.240)}$
 $= 7.96 \approx 8 \text{ Hz}$

C1

B1

C1

B1

A0 [4]

- (b) (i) resonance

B1 [1]

- (ii) 8 Hz

B1 [1]

- (c) (increase amount of) damping
 without altering (k or) m ... (some indirect reference is acceptable)
 sensible suggestion

B1

B1

B1 [3]

- 4 (a) (i) $GM = (R \times 10^3) = 1 \times (R \times 10^3) = 1$

B1